



An acrylated fullerene derivative for efficient and thermally stable polymer solar cells



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ABSTRACT

The continuous microstructure evolution occurring in active layers of polymer-fullerene solar cells is one of the main causes for their device instability. With aim to tackle it, this work developed a new polymerizable fullerene acceptor, [6,6]-phenyl-C₆₁-butyl acrylate (PC₆₁BA). It was found that PC₆₁BA has similar light-absorption properties and HOMO and LUMO energy levels as [6,6]-phenyl-C₆₁-butyric acid methyl ester (PC₆₁BM), and can be converted into insoluble oligomers upon heating at 150 °C. Polymer-fullerene solar cells using poly(3-hexylthiophene) (P3HT) as donor and PC₆₁BA as acceptor exhibited an optimized efficiency of 3.54%, the performance comparable to P3HT/PC₆₁BM cells (optimized efficiency: 3.70%). But, the former possess much better thermal stability than the latter owing to aggregation suppression by the polymerized PC₆₁BA. These results indicate that PC₆₁BA, unlike most previous reported, is a unique polymerizable fullerene derivative that can be used alone as acceptor to achieve both efficient and thermally stable polymer solar cells.

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Introduction

Bulk heterojunction (BHJ) polymer solar cells (PSCs) have attracted considerable attention owing to their advantages of light weight, flexibility, adaptability to low cost solution processing and large area device fabrication. With great success in material innovation and device engineering in the past decades,¹ PSCs have achieved a significant progress with power conversion efficiency (PCE) record raised from 3–5% in 2000s^{1,2} to nowadays over 10%.^{1,3} Although such efficiency level is needed to be further improved for their real use, it is appropriate to pay attention on device stability issue at the same time.⁴

According to light-to-electric conversion mechanism, a BHJ active layer is generally composed of a physical blend of electron donor (D) and acceptor (A) materials.¹ In order to provide a large D/A interface for efficient exciton dissociation (or charge separation)

and meanwhile the transportation pathways for both hole and electron charge carriers, a bi-continuous interpenetrating D/A biphasic microstructure should be formed in the active layer. Moreover, their domain widths are better to keep in the same magnitude of exciton effective diffusion length (10–20 nm) for a high exciton migration efficiency. Although it is still known little about what governing microstructure formation for a physical blend of two materials and how to precisely control their formation, a number of PSC systems have achieved nearly ideal configuration after careful optimization on device fabrication parameters and procedures, including donor/acceptor ratio, processing solvent and temperature, additives, concentration, spin-coating rate, thermal annealing, solvent vapor annealing, standing alone, and so on, and thus endowed them with high performance.^{1–3} Since no specific factors and interactions are engaged to hold blend components still, the so-formed nanoscale phase-separated microstructures are not thermally stable, and absolutely continue to evolve into large phase separation. And even in many reported cases, large aggregates were clearly naked identified when their active layers were subjected to thermal treatment.⁵ Therefore, such microstructure evolution occurring in PSC active layers mostly leads to reduction on the size of D/A interface, destruction on the hole and electron

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